



ALBIN RAJESH

SOFTWARE DEVELOPER

CONTACT

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EDUCATION

2022 - 2026

AMAL JYOTHI COLLEGE OF ENGINEERING [AUTONOMOUS]

- B-Tech in Information Technology.

2020 - 2022

CITADEL RESIDENTIAL SCHOOL, RANNI

- Higher Secondary Education
- CBSE

TECHNICAL SKILLS

- Languages** : Python, Java, C
- Web & Backend** : HTML5, CSS3, Django, REST APIs, Node.js, Express.js
- Databases** : SQL, PostgreSQL, MongoDB, Firebase
- Systems & Tools** : Docker, Git, Raspberry Pi, Arduino IDE, Linux (Ubuntu)
- AI & Signal Processing** : NumPy, SciPy, FFT, Spectrogram Analysis, rPPG, Machine Learning, Deep Learning, PyTorch, OpenCV

PROFILE

Versatile Software Developer skilled in embedded systems, audio signal processing, AI, and full-stack web development. Experienced in IoT automation, Raspberry-Pi optimization, and scalable Django applications, delivering practical hardware-integrated solutions.

WORK EXPERIENCE

AMM Labs

Python Developer Intern

Jan 2025 - Feb 2025

- Built and optimized a real-time acoustic fault detection system for Raspberry Pi using audio signal processing.
- Real-time mechanical fault detection from sound using FFT-based audio fingerprinting optimized for embedded systems.
- Technologies: Python, NumPy, FFT, Spectrograms, Raspberry Pi

Accelerate X & Novolo AI

Artificial Intelligence Intern

Jun 2025 - Jul 2025

- Built deep learning models (CNNs, 1D ResNet) for high-accuracy image and pattern recognition.
- Improved classification metrics and model latency by optimizing ML/DL training and preprocessing workflows.
- Technologies: Python, PyTorch, CNN, ResNet-1D, Machine Learning, Deep Learning, NLP

PROJECTS

VitalSense

Camera Based Heart Rate Monitoring System

- Engineered a continuous remote Photoplethysmography (rPPG) pipeline estimating HR, HRV, and stress using hybrid POS and deep learning consensus models.
- Led a team to develop and submit this real-time health monitoring architecture to the institutional I2U contest.
- Technologies: Python, OpenCV, rPPG, TS-CAN, ResNet-1D, LightGBM

NativeGoods

Local Seller E-Commerce Platform

- Created a scalable platform enabling local sellers to list, manage, and process products online.
- Designed a responsive frontend interface and integrated PostgreSQL for secure database management and query optimization.
- Tools: Django, PostgreSQL, HTML, CSS, JS